

Applied NLP

DATA 641

Lecture 4 — Text Representation; Bag of N-Grams

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Bag of N-Grams

- All the representation schemes we have seen so far treat words as independent units. There is no notion of phrases or word ordering. The bag-of-n-grams (BoN) approach tries to remedy this. It does so by breaking text into chunks of n contiguous words (or tokens).
- This can help us capture some context, which earlier approaches could not do. Each chunk is called an n -gram.
- The corpus vocabulary, V , is a collection of all unique n -grams across the text corpus. Then, each document in the corpus is represented by a vector of length $|V|$. This vector simply contains the frequency counts of n -grams present in the document and zero for the n -grams that are not present.
- The set of all bigrams in the corpus is as follows: {dog bites, bites man, man bites, bites dog, dog eats, eats meat, man eats, eats food}. Then, BoN representation consists of an eight-dimensional vector for each document. The bigram representation for the first two documents is as follows: D1 : [1,1,0,0,0,0,0,0] , D2 : [0,0,1,1,0,0,0,0].
- Note that, by increasing the value of n , we can incorporate larger context; however, this further increases the sparsity. In NLP parlance, the BoN scheme is also called "n-gram feature selection."

Some of the main **advantages** and **disadvantages** of BoN include:

- BoN captures some context and word-order information in the form of n-grams.
- Thus, resulting vector space is able to capture some semantic similarity. Documents having the same n-grams will have their vectors closer to each other in Euclidean space as compared to documents with completely different n-grams.
- As n increases, dimensionality (and therefore sparsity) only increases rapidly.
- It still provides no way to address the OOV problem.